

HORNSBY GIRLS HIGH SCHOOL



2011 TRIAL HIGHER SCHOOL CERTIFICATE EXAMINATION

Mathematics Extension 2

General Instructions

- Reading Time – 5 minutes
- Working Time – 3 hours
- Write using a black or blue pen
- Approved calculators may be used
- A table of standard integrals is provided at the back of this paper
- All necessary working should be shown for every question
- Begin each question in a new booklet

Total marks (120)

- Attempt Questions 1 – 8
- All questions are of equal value

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Total Marks**Attempt Questions 1–8****All Questions are of equal value**

Begin each question in a new booklet, writing your student number and question number in the boxes indicated. Extra writing booklets are available.

Question 1 (15 marks) Use a SEPARATE writing booklet. **Marks**

(a) Find $\int \frac{x^2}{\sqrt{8-x^3}} dx$. 2

(b) By completing the square, find $\int \frac{dx}{x^2-8x+20}$. 2

(c) Evaluate $\int_0^\pi x \cos x \, dx$. 3

(d) (i) Show that $\frac{2}{x^3+x^2+x+1} = \frac{1}{x+1} - \frac{x}{x^2+1} + \frac{1}{x^2+1}$. 2

(ii) Hence, or otherwise, show $\int_{\frac{1}{2}}^2 \frac{2}{x^3+x^2+x+1} = \tan^{-1} 2 - \tan^{-1} \left(\frac{1}{2} \right)$. 3

(e) Using the substitution $x = \tan \theta$, or otherwise, show $\int_1^{\sqrt{3}} \frac{1}{x^2 \sqrt{1+x^2}} dx = \sqrt{2} - \frac{2}{\sqrt{3}}$. 3

Question 2 (15 marks) Use a SEPARATE writing booklet.

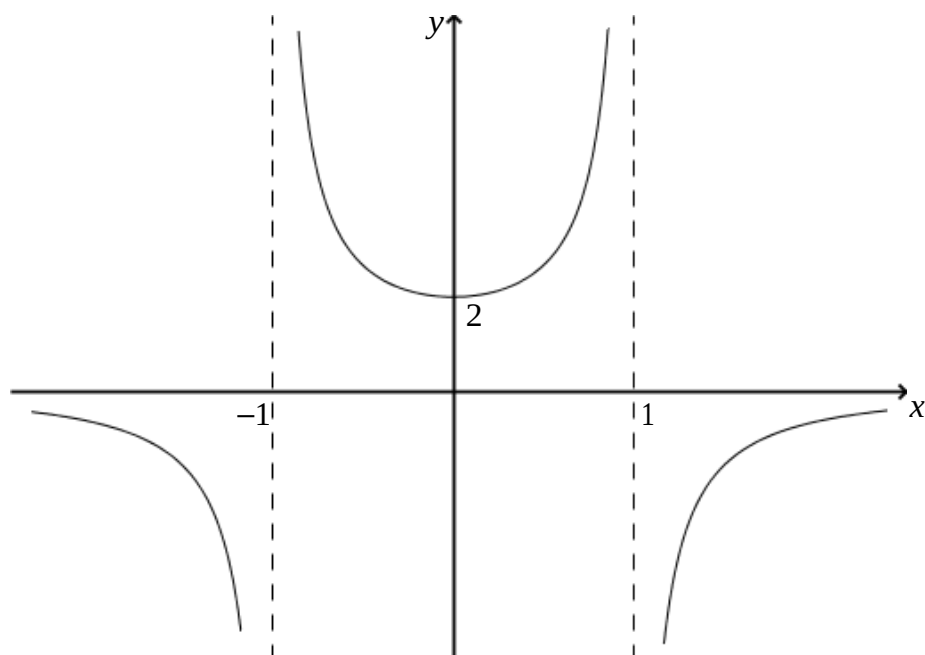
Marks

- (a) Write i^7 in the form $x + iy$ where x and y are real.
1
- (b) Let $z = 2 + 2i$ and $w = 2 - i$. Find in the form $x + iy$, where x and y are real,
- (i) $z\bar{w}$ 1
- (ii) $\frac{8}{z}$ 1
- (c) It is given that $1 + i$ is a root of $P(z) = 2z^3 - 3z^2 + rz + s$, where r and s are real.
- (i) Explain why $1 - i$ is also a root of the equation. 1
- (ii) Factorise $P(z)$ over the real field. 2
- (d) Find all the solutions of $z^4 = 16$. Express your solutions in the modulus-argument form. 2
- (e) Sketch the region in the complex plane where the inequalities $|z - \bar{z}| \leq 2$ and $|z - i| \leq 4$ hold. 3
- (f) (i) Prove, by Mathematical Induction, that for all integers n ,
 $(\cos \theta + i \sin \theta)^n = \cos(n\theta) + i \sin(n\theta)$ 3
- (ii) Hence, find an expression for $\cos 3\theta$. 1

Question 3 (15 marks) Use a SEPARATE writing booklet.

Marks

- (a) The diagram shows the graph of $y = f(x)$. The graph has a horizontal asymptote at $y = 0$ and vertical asymptotes at $x = \pm 1$.



NOT TO SCALE

Draw neat separate one-third page sketches of the graphs of the following:

(i) $y = \frac{1}{f(x)}$ 2

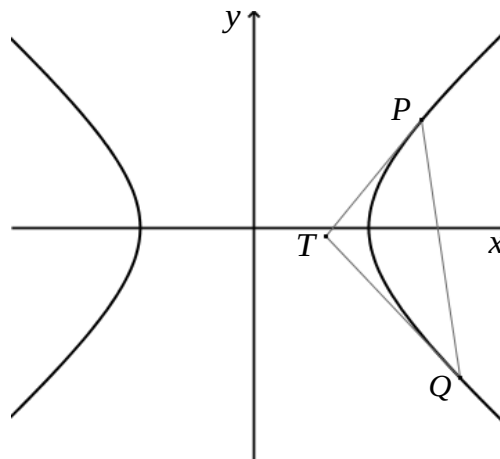
(ii) $y = f(x) + |f(x)|$ 2

(iii) $y = e^{f(x)}$ 2

Question 3 continues on page 6

Question 3 (continued)

(b)



NOT TO SCALE

The points $P(x_1, y_1)$ and $Q(x_2, y_2)$ lie on the right branch of the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$.

The tangents at P and Q meet at $T(x_0, y_0)$.

(i) Show the equation of the tangent at P is $\frac{xx_1}{a^2} - \frac{yy_1}{b^2} = 1$ 2

(ii) Hence show the equation of the chord of contact is $\frac{xx_0}{a^2} - \frac{yy_0}{b^2} = 1$. 2

(iii) The chord PQ passes through the focus $S(ae, 0)$ where e is the eccentricity 1
of the hyperbola. Prove T lies on the directrix of the parabola.

(c) Let α, β, γ be the zeros of the polynomial $P(x) = 3x^3 + 7x^2 + 11x + 51$.

(i) Find $\alpha^2\beta\gamma + \alpha\beta^2\gamma + \alpha\beta\gamma^2$. 1

(ii) Find $\alpha^2 + \beta^2 + \gamma^2$. 2

(iii) Using part (ii), or otherwise, determine how many zeros of $P(x)$ are real. 1
Justify your answer.

Question 4 (15 marks) Use a SEPARATE writing booklet.

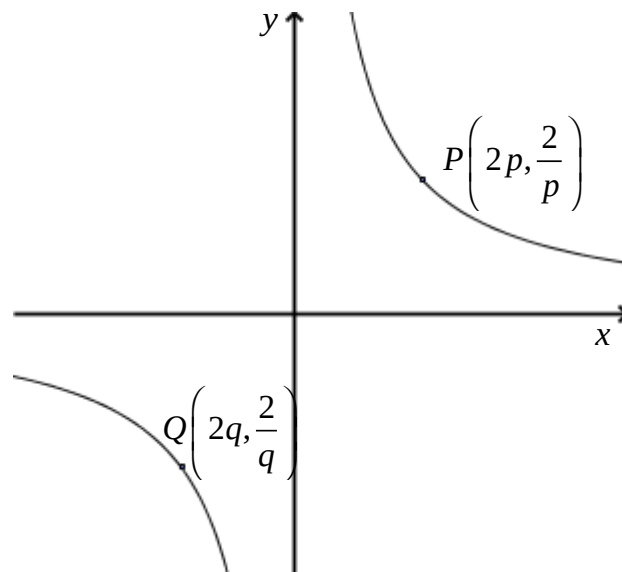
Marks

- (a) A solid of height 2 metres rests on a horizontal surface. 3

Every horizontal cross-section of the solid, x metres above the surface, is a square of side $\sqrt{3x+1}$ metres.

Find the volume of the solid.

- (b) Consider the rectangular hyperbola $xy = 4$, with points P and Q on different branches of the hyperbola.



NOT TO SCALE

- (i) Prove that the equation of the normal to $xy = 4$ at the point $P\left(2p, \frac{2}{p}\right)$ is $py - p^3x = 2(1 - p^4)$. 3

- (ii) If this normal meets the hyperbola again at $Q\left(2q, \frac{2}{q}\right)$, prove that $q = \frac{-1}{p^3}$. 2

- (iii) Hence, show that there exists only one chord of the hyperbola which is normal to the hyperbola at P and Q , and find its equation. 3

- (c) The equation $x^3 + 3x + 2 = 0$ has roots α , β and γ .

- (i) Find the polynomial whose roots are α^2 , β^2 and γ^2 . 2

- (ii) Hence, or otherwise, find the value of $\alpha^3 + \beta^3 + \gamma^3$. 2

Question 5 (15 marks) Use a SEPARATE writing booklet.

Marks

(a) Let w be a complex root of unity (w is solution of $z^3 - 1 = 0$).

(i) Show that $(z-1)(z^2+z+1) = z^3-1$. 1

(ii) Explain why $w^2 + w + 1 = 0$. 1

(iii) Hence, or otherwise, show that $(1-w)(1-w^2)(1-w^4)(1-w^8) = 9$ 3

(b) Consider $I = \int_1^{\infty} \frac{1}{x\sqrt{1+x^2}} dx$.

(i) By using a suitable substitution, show that $I = \int_{\frac{\pi}{4}}^{\frac{\pi}{2}} \frac{1}{\sin \theta} d\theta$. 2

(ii) Hence, or otherwise, evaluate I . 3

(c) (i) Find real numbers, a and b , such that 2

$$x^4 + x^3 + x^2 + x + 1 = (x^2 + ax + 1)(x^2 + bx + 1).$$

(ii) Given that $x = \cos \frac{2\pi}{5} + i \sin \frac{2\pi}{5}$ is a solution of $x^4 + x^3 + x^2 + x + 1 = 0$,

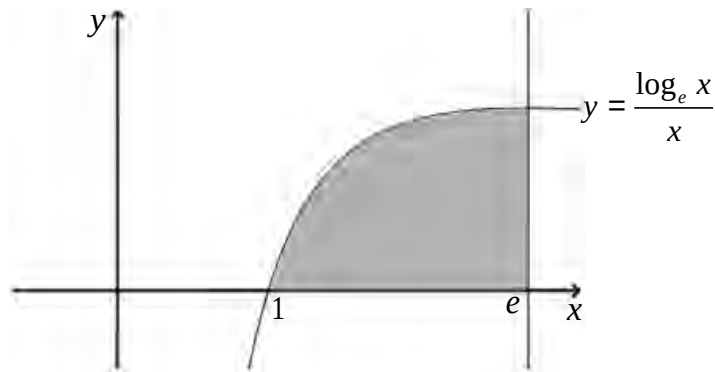
find the exact value of $\cos \frac{2\pi}{5}$. 3

Question 6 (15 marks) Use a SEPERATE writing booklet.**Marks**

- (a) Use the method of cylindrical shells to find the volume of the solid 4

formed when the shaded region bounded by $y = 0$, $y = \frac{\log_e x}{x}$ and $x = e$

is rotated about the y-axis.



NOT TO SCALE

- (b) (i) Show that $\sin(\alpha + \beta) + \sin(\alpha - \beta) = 2 \sin \alpha \cos \beta$. 1

- (ii) Hence, or otherwise, solve the equation 3

$$\sin \theta + \sin 2\theta + \sin 3\theta + \sin 4\theta = 0 \text{ for } 0 \leq \theta \leq 2\pi.$$

- (c) A stone is projected vertically upwards in the air from a point h metres above 3

the ground at a speed u and experiences a resistance equal to mkv^2 , where m is the mass of the stone, v is the speed after time t and k is a constant.

By considering the forces acting on the stone, show that the maximum height, H ,

the stone reaches above the ground is given by $H = h + \frac{1}{2k} \ln \left(1 + \frac{ku^2}{g} \right)$, where g is

acceleration due to gravity

- (d) A group of n people are to be seated around a circular table. Find the number 2

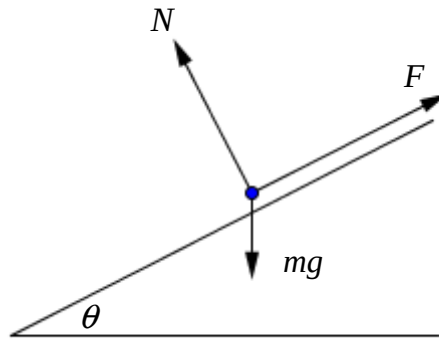
of possible arrangements if 3 particular people are to sit together.

- (e) Show that ${}^{n+2}C_r = {}^nC_r + 2{}^nC_{r-1} + {}^nC_{r-2}$ 2

Question 7 (12 marks) Use a SEPARATE writing booklet.

Marks

(a)



A particle of mass m is lying on an inclined plane and does not move.

2

The plane is at an angle θ to the horizontal. The particle is subject to a gravitational force mg , a normal reaction force N , and a frictional force F parallel to the plane, as shown in the diagram above.

By resolving the forces acting on the particle parallel and perpendicular to the plane, find an expression for $\frac{F}{N}$ in terms of θ .

(b) The polynomial $P(x) = x^4 - 4x^3 + 3x^2 - 14x + 10$ has roots $a + ib$, $a - 2ib$, where a and b are real.

(i) Show that $a = 1$, and hence find the value(s) of b .

2

(ii) Hence, factorise $P(x)$ over the rational field.

2

(c) (i) If $I_n = \int_0^{\frac{\pi}{2}} \cos^n x \, dx$, then show that $I_n = \frac{n-1}{n} I_{n-2}$.

3

(ii) Hence, evaluate $\int_0^{\frac{\pi}{2}} \cos^5 x \, dx$.

2

(d) Use Mathematical Induction to prove that for integer values of $n \geq 1$
 $1 \times 1! + 2 \times 2! + 3 \times 3! + \dots + n \times n! = (n+1)! - 1$

4

Question 8 (15 marks) Use a SEPARATE writing booklet.

Marks

(a) The function $y = f(x)$ is defined by $f(x) = \sqrt{3 - \sqrt{x}}$

(i) State the domain of the function $f(x)$. **1**

(ii) Show that $y = f(x)$ is a decreasing function and determine the range of $y = f(x)$. **2**

(iii) Sketch the graph of $y = f(x)$ for the domain and range determined above. **1**

(iv) Prove that $\int_0^9 \sqrt{3 - \sqrt{x}} dx = \frac{24\sqrt{3}}{5}$ **2**

(b) Show that $\frac{d}{du}(\sec u + \tan u) = \sec u(\sec u + \tan u)$ **2**

(c) Consider $f(x) = \cos \frac{x}{2}$.

(i) On the same set of axes, sketch the graph of $y = f(x)$, and hence the graph of $y = \frac{1}{f(x)}$ for the domain $0 \leq x \leq 2\pi$. **2**

(ii) By considering part (b), find the area bounded by the curve $y = \frac{1}{f(x)}$, the x -axis and the ordinates $x = \frac{\pi}{3}$ and $x = \frac{\pi}{2}$, leaving your answer exact. **3**

(iii) The solid bounded by the curve $y = \frac{1}{f(x)}$, the x -axis and the ordinates $x = \frac{\pi}{3}$ and $x = \frac{\pi}{2}$ is rotated about the y -axis. **2**

By using the method of annular discs, find the volume as a **definite integral**.
DO NOT EVALUATE THIS INTEGRAL.

End of paper

Question 1

$$(a) \int \frac{x^2}{\sqrt{8-x^3}} dx = \frac{1}{3} \int \frac{du}{u^{1/2}} \quad \begin{array}{l} u = 8-x^3 \\ u' = -3x^2 \end{array}$$

$$= -\frac{1}{3} [2u^{1/2}] + C$$

$$= -\frac{2}{3} \sqrt{8-x^3} + C$$

$$(b) \int \frac{dx}{x^2-8x+20} = \int \frac{dx}{(x-4)^2+4}$$

$$= \frac{1}{2} \tan^{-1} \left(\frac{x-4}{2} \right) + C$$

$$(c) \int_0^{\pi} x \cos x dx$$

$$= [x \sin x]_0^{\pi} - \int_0^{\pi} \sin x dx$$

$$= 0 - [-\cos x]_0^{\pi}$$

$$= 0 - [1 - 1]$$

$$= -2$$

$u = x$
 $du = dx$
 $dv = \cos x dx$
 $v = \sin x$

(d)(i) RTS $\frac{2}{x^3+x^2+x+1} = \frac{1}{x+1} - \frac{x}{x^2+1} + \frac{1}{x^2+1}$

$$RHS = \frac{1}{x+1} - \frac{x}{x^2+1} + \frac{1}{x^2+1}$$

$$= \frac{x^2+1-x^2-x+1}{(x+1)(x^2+1)}$$

$$= \frac{2}{x^3+x^2+x+1}$$

= LHS as req.

(ii) $\int_{\frac{1}{2}}^2 \frac{2}{x^3+x^2+x+1} dx = \int_{\frac{1}{2}}^2 \left(\frac{1}{x+1} - \frac{x}{x^2+1} + \frac{1}{x^2+1} \right) dx$

$$= \left[\ln(x+1) - \frac{1}{2} \ln(x^2+1) + \tan^{-1} x \right]_{\frac{1}{2}}^2$$

(d)(ii)(cont.)

$$= \left[\ln 3 - \frac{1}{2} \ln 5 + \tan^{-1} 2 \right]$$

$$- \left[\ln \frac{3}{2} - \frac{1}{2} \ln \frac{5}{4} + \tan^{-1} \frac{1}{2} \right]$$

$$= \ln \left[3 \times \frac{1}{5} \times \frac{2}{3} \times \frac{5}{2} \right] + \tan^{-1} 2 + \tan^{-1} \frac{1}{2}$$

$$= \ln 1 + \tan^{-1} 2 + \tan^{-1} \frac{1}{2}$$

$$= \tan^{-1} 2 + \tan^{-1} \frac{1}{2} \text{ as req.}$$

(e) $\int_1^{\sqrt{3}} \frac{dx}{x^2 \sqrt{1+x^2}}$

$x = \tan \theta$
 $dx = \sec^2 \theta d\theta$
 $x = \sqrt{3} \Rightarrow \theta = \frac{\pi}{3}$
 $x = 1 \Rightarrow \theta = \frac{\pi}{4}$

$$= \int_{\frac{\pi}{4}}^{\frac{\pi}{3}} \frac{\sec^2 \theta d\theta}{\tan^2 \theta \sqrt{1+\tan^2 \theta}}$$

$$= \int_{\frac{\pi}{4}}^{\frac{\pi}{3}} \frac{\sec^2 \theta d\theta}{\frac{\sin^2 \theta}{\cos^2 \theta} \cdot \sec \theta}$$

$$= \int_{\frac{\pi}{4}}^{\frac{\pi}{3}} \cos \sec \theta \cdot \cot \theta d\theta$$

$$= - [\csc \theta]_{\frac{\pi}{4}}^{\frac{\pi}{3}}$$

$$= - \left(\frac{2}{\sqrt{3}} - \sqrt{2} \right)$$

$$= \sqrt{2} - \frac{2}{\sqrt{3}}$$

Question 2.

$$\begin{aligned} a) \quad i^7 &= i^2 i^2 i^2 i \\ &= (-1) \cdot (-1) \cdot (-1) \cdot i \\ &= -i \\ &= 0 - i \end{aligned}$$

$$\begin{aligned} b) \quad z &= 2 + 2i \\ w &= 2 - i \end{aligned}$$

$$\begin{aligned} i) \quad z\bar{w} &= (2+2i)(2-i) \\ &= 4 + 2i + 4i + 2i^2 \\ &= 2 + 6i \end{aligned}$$

$$\begin{aligned} ii) \quad \frac{z}{z} &= \frac{z\bar{z}}{z\bar{z}} \\ &= \frac{z(2-2i)}{(2+2i)(2-2i)} \\ &= \frac{z(2-2i)}{2^2+2^2} \\ &= 2-2i \end{aligned}$$

c) $P(z)$ has real coefficients, hence roots occur in conjugate pairs

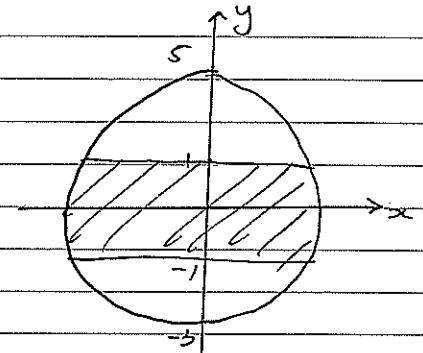
$$\begin{aligned} i) \quad 1+i+1-i+a &= \frac{3}{2} \\ a &= -\frac{1}{2} \end{aligned}$$

$$\begin{aligned} \therefore P(z) &= 2(z - (1+i))(z - (1-i))(z + \frac{1}{2}) \\ &= (z^2 - 2z + 2)(2z + 1) \end{aligned}$$

$$\begin{aligned} d) \quad z^4 &= 16 \\ (z^2+4)(z^2-4) &= 0 \\ z &= 2, -2 \\ z^2 &= -4 \\ z &= \pm\sqrt{-4} \\ z &= \pm 2i \\ &= \pm 2i \end{aligned}$$

$$\therefore z = 2, -2, 2i, -2i$$

$$\begin{aligned} e) \quad |z-i| &\leq 4 \\ \text{circle centre } (0, 1) \\ \text{radius } 4. \\ |x+iy - (x-iy)| &\leq 2 \\ |2iy| &\leq 2 \\ -1 &\leq y \leq 1 \end{aligned}$$



$$\begin{aligned} f) \quad (\cos\theta + i\sin\theta)^n &= \cos n\theta + i\sin n\theta \\ \text{Test } n=0 \\ \text{LHS} &= (\cos\theta + i\sin\theta)^0 \\ &= 1 \\ \text{RHS} &= \cos 0 + i\sin 0 \\ &= 1 \end{aligned}$$

$$\begin{aligned} \text{Assume true } n=k: \\ (\cos k\theta + i\sin k\theta)^n &= \cos n\theta + i\sin n\theta \end{aligned}$$

$$f) i) (\cos \theta + i \sin \theta)^n = \cos(n\theta) + i \sin(n\theta)$$

test $n=1$.

$$\text{LHS} = \cos \theta + i \sin \theta$$

$$\text{RHS} = \cos \theta + i \sin \theta$$

$$\text{LHS} = \text{RHS}$$

$\therefore$ true for $n=1$.

assume true for $n=k$

$$\text{i.e. } (\cos \theta + i \sin \theta)^k = \cos(k\theta) + i \sin(k\theta)$$

Prove true for $n=k+1$

$$\text{RTP: } (\cos \theta + i \sin \theta)^{k+1} = \cos(k+1)\theta + i \sin(k+1)\theta$$

$$(\cos \theta + i \sin \theta)^{k+1} = (\cos \theta + i \sin \theta)(\cos \theta + i \sin \theta)^k$$

$$= (\cos \theta + i \sin \theta)(\cos k\theta + i \sin k\theta)$$

$$= \cos \theta \cos k\theta + i \sin k\theta \cos \theta + i \sin \theta \cos k\theta$$

$$- \sin \theta \sin k\theta$$

$$= \cos \theta \cos k\theta - \sin \theta \sin k\theta + i(\sin k\theta \cos \theta + \sin \theta \cos k\theta)$$

$$= \cos(\theta + k\theta) + i \sin(\theta + k\theta)$$

$$= \cos \theta(k+1) + i \sin \theta(k+1)$$

$\therefore$ if true for $n=1$, it must be true for $n=2, n=3$ and all other positive integers where $n \geq 1$.

test $n=0$.

$$\text{LHS} = 1$$

$$\text{RHS} = 1$$

$\therefore$ true for $n=0$.

$\rightarrow$

Start here.

test for negative integers

$$\text{Pr. } (\cos \theta + i \sin \theta)^{-n} = \cos(-n\theta) + i \sin(-n\theta)$$

$$(\cos \theta + i \sin \theta)^{-n} = ((\cos \theta + i \sin \theta)^n)^{-1}$$

$$= (\cos n\theta + i \sin n\theta)^{-1}$$

$$= \frac{1}{z}$$

$$\text{where } z = \cos n\theta + i \sin n\theta$$

$$= \frac{\overline{z}}{z\overline{z}}$$

$$= \frac{\overline{z}}{|z|^2}$$

$$= \cos n\theta - i \sin n\theta$$

$$= \cos(-n\theta) + i \sin(-n\theta)$$

$\therefore$ true for all negative integers

$\therefore (\cos \theta + i \sin \theta)^n = \cos(n\theta) + i \sin(n\theta)$ true for all integers

ii) ~~test~~

$$(\cos \theta + i \sin \theta)^3 = (\cos 3\theta + i \sin 3\theta)$$

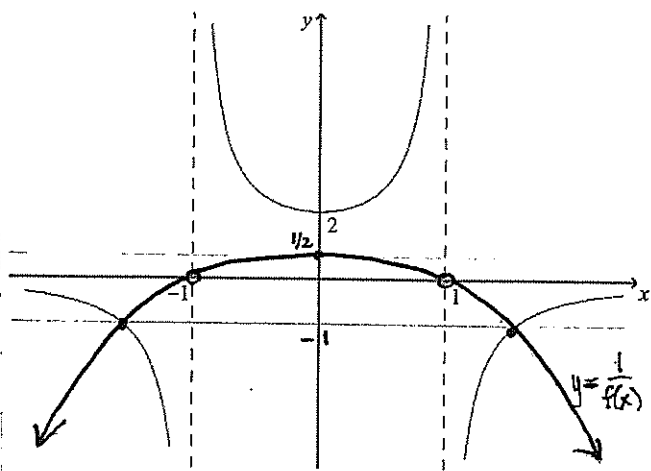
$$\cos^3 \theta + i 3 \cos^2 \theta \sin \theta - 3 \cos \theta \sin^2 \theta - i \sin^3 \theta$$

equating real part.

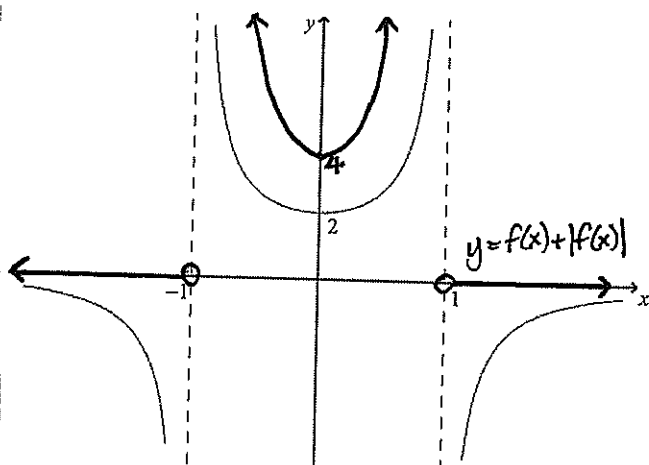
$$\cos 3\theta = \cos^3 \theta - 3 \cos \theta \sin^2 \theta$$

Question 3

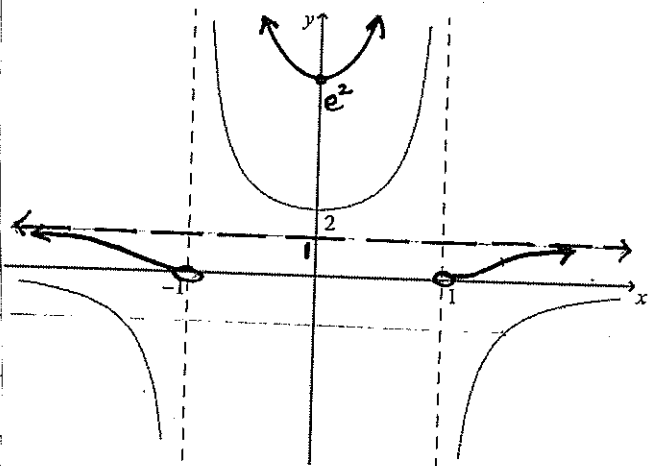
(a) (i)



(ii)



(iii)



(b) (i) $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$

$\therefore \frac{2x}{a^2} - \frac{2y}{b^2} \frac{dy}{dx} = 0$

$-\frac{2y}{b^2} \frac{dy}{dx} = -\frac{2x}{a^2}$

$\frac{dy}{dx} = \frac{b^2 x}{a^2 y}$

b(i) (cont.) $\therefore$ gradient at $P(x_1, y_1) = \frac{b^2 x_1}{a^2 y_1}$

$\therefore$ equation of tangent at P

$$y - y_1 = \frac{b^2 x_1}{a^2 y_1} (x - x_1)$$

$$a^2 y y_1 - a^2 y_1^2 = b^2 x_1 x - b^2 x_1^2$$

$$\therefore b^2 x_1 x - a^2 y_1 y = b^2 x_1^2 - a^2 y_1^2$$

$$\frac{x_1 x}{a^2} - \frac{y_1 y}{b^2} = \frac{x_1^2}{a^2} - \frac{y_1^2}{b^2}$$

$$\frac{x_1 x}{a^2} - \frac{y_1 y}{b^2} = 1 \quad \text{since } (x_1, y_1) \text{ lies on the hyperbola.}$$

(ii) Eq. of tangent at P $\frac{x_1 x}{a^2} - \frac{y_1 y}{b^2} = 1$

sim. Eq. of tangent at Q $\frac{x_2 x}{a^2} - \frac{y_2 y}{b^2} = 1$

T(x_0, y_0) lies on both

$$\therefore \frac{x_1 x_0}{a^2} - \frac{y_1 y_0}{b^2} = 1 \quad \text{P thru' T}$$

$$\frac{x_2 x_0}{a^2} - \frac{y_2 y_0}{b^2} = 1 \quad \text{Q thru' T}$$

$\therefore$ Eq. of PQ (chord of contact) is

$$\frac{x x_0}{a^2} - \frac{y y_0}{b^2} = 1 \quad \text{as } P(x_1, y_1) \text{ and } Q(x_2, y_2) \text{ both}$$

satisfy this equation.

(iii) If PQ passes thru' S($ae, 0$) then

$$\frac{x_0 ae}{a^2} - \frac{y(0)}{b^2} = 1$$

$$x_0 = \frac{a}{e}$$

$\therefore$ T lies on the directrix

Question 3

(c) $P(x) = 3x^3 + 7x^2 + 11x + 5$

$$\begin{aligned} \text{(i)} \quad \alpha^2\beta\gamma + \alpha\beta^2\gamma + \alpha\beta\gamma^2 &= \alpha\beta\gamma(\alpha + \beta + \gamma) \\ &= -\frac{5}{3} \times -\frac{7}{3} \\ &= \frac{119}{9} \text{ OR } 39\frac{2}{3} \end{aligned}$$

$$\begin{aligned} \text{(ii)} \quad \alpha^2 + \beta^2 + \gamma^2 &= (\alpha + \beta + \gamma)^2 - 2(\alpha\beta + \alpha\gamma + \beta\gamma) \\ &= \left(-\frac{7}{3}\right)^2 - 2 \times \frac{11}{3} \\ &= \frac{49}{9} - \frac{22}{3} \\ &= -\frac{17}{9} \text{ OR } -1\frac{8}{9} \end{aligned}$$

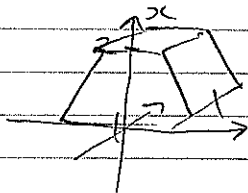
(iii) $\alpha^2 + \beta^2 + \gamma^2 < 0 \therefore$ at least one root is unreal.

i.e. at least α^2 or β^2 or $\gamma^2 < 0$
also as the coefficients are all real, the roots occur in conjugates so if one root is unreal its conjugate is also unreal.

As $P(x)$ is degree 3 there is then only one more zero which must be real. Therefore there is exactly one root of $P(x)$ that is real.

Question 4.

(a)



$$\text{Area} = (\sqrt{3x+1})^2$$

$$= 3x+1.$$

$$V = \lim_{\Delta x \rightarrow 0} \sum_{x=0}^2 (3x+1) \Delta x$$

$$= \int_0^2 (3x+1) dx$$

$$= \left[\frac{3x^2}{2} + x \right]_0^2$$

$$= \frac{3}{2}(2)^2 + 2$$

$$= 8 \text{ m}^3.$$

(b)

$$(i) y = \frac{4}{x}$$

$$\frac{dy}{dx} = -\frac{4}{x^2}$$

At P ,

$$\frac{dy}{dx} = -\frac{1}{p^2}.$$

$\therefore$ gradient of normal is p^2

$$y - \frac{2}{p} = p^2(x - 2p)$$

$$py - 2 = p^3x - 2p^4$$

$$py - p^3x = 2(1 - p^4).$$

$$(ii) py - p^3x = 2 - 2p^4$$

$$\text{sub } Q(2q, \frac{2}{q})$$

$$p \cdot \frac{2}{q} - p^3 \cdot 2q = 2 - 2p^4$$

$$2p - 2p^3q^2 = 2q - 2qp^4$$

$$p - p^3q^2 - q + qp^4 = 0$$

$$qp^4 - p^3q^2 + p - q = 0$$

$$qp^3(p - q) + (p - q) = 0$$

$$\therefore qp^3 + 1 = 0 \quad (p \neq q)$$

$$qp^3 = -1$$

$$q = -\frac{1}{p^3}.$$

$$(ii) \quad q = \frac{-1}{p^3}$$

$$p = \frac{-1}{q^3}$$

$$\therefore pq^3 = qp^3$$

$$pq^3 - qp^3 = 0$$

$$pq(q^2 - p^2) = 0$$

$$q^2 - p^2 = 0$$

$$(q-p)(q+p) = 0$$

$$q = p, \quad q = -p.$$

$$\therefore \text{When } q = p$$

$$p^4 = -1$$

No sol.

$$q = -p$$

$$-p^4 = -1$$

$$p = \pm 1$$

$$\text{Sub } p = 1,$$

$$y - x = 0$$

$$y = x$$

$$\text{Sub } p = -1$$

$$-y + x = 0$$

$$y = x$$

$$\therefore \text{Normal is } y = x$$

Start here.

$$c) \quad x^3 + 3x + 2 = 0.$$

$$y = x^2, \quad x = \alpha, \beta, \gamma.$$

$$\sqrt{y} = x.$$

$$(\sqrt{x})^3 + 3(\sqrt{x}) + 2 = 0.$$

$$x\sqrt{x} + 3\sqrt{x} + 2 = 0.$$

$$\sqrt{x}(x+3) = -2.$$

$$x^2(x^2+6x+9) = 4.$$

$$\cancel{x^2 + 6x + 9 = 0}$$

$$x^3 + 6x^2 + 9x - 4 = 0.$$

$$b) \quad ii) \quad \alpha^3 + \beta^3 + \gamma^3 \quad \text{from } x^3 + 3x + 2$$

$$\alpha^3 + 3\alpha + 2 = 0. \quad ①$$

$$\beta^3 + 3\beta + 2 = 0. \quad ②$$

$$\gamma^3 + 3\gamma + 2 = 0. \quad ③$$

2

$$① + ② + ③$$

$$\alpha^3 + \beta^3 + \gamma^3 = -3(\alpha + \beta + \gamma) - 2 \cdot 3.$$

$$= -3(0) - 6.$$

$$= -6.$$

Question 5

a) RS: $(z-1)(z^2+z+1) = z^3-1$

$$\begin{aligned} \text{LHS} &= (z-1)(z^2+z+1) \\ &= z^3+z^2+z-z^2-z-1 \\ &= z^3-1 \end{aligned}$$

ii) w is a complex root of unity

$$w^3-1=0$$

$$\therefore (w-1)(w^2+w+1)=0$$

But $w \neq 1$, $w \in \mathbb{C}$

$$\therefore w^2+w+1=0$$

ii) RS: $(1-w)(1-w^2)(1-w^4)(1-w^8)=9$

$$\text{LHS} = (1-w)(1-w^2)(1-w^4)(1-w^8)$$

$$w^4 = w^3 \cdot w$$

$$= w$$

$$w^8 = w^3 \cdot w^3 \cdot w^2$$

$$= w^2$$

$$w^2+w+1=0$$

$$w^2 = -w-1$$

$$w+1 = -w^2$$

$$\therefore \text{LHS} = (1-w)^2(1-w^2)^2$$

$$= (1-2w+w^2)(1-2w^2+w^4)$$

$$= (1-2w-w-1)(1-2(-w-1)+w)$$

$$= (-3w)(1+2w+2+w)$$

$$= (-3w)(3+3w)$$

$$= -3w(-3w^2)$$

$$= 9w^3$$

$$= 9$$

(b) i) $I = \int_1^{\infty} \frac{1}{x\sqrt{1+x^2}} dx$

let $x = \tan \theta$

when $x=1$, $\theta = \pi/4$

$$\frac{dx}{d\theta} = \sec^2 \theta$$

$x = \infty$, $\theta = \pi/2$

$$dx = \sec^2 \theta d\theta$$

$$\therefore I = \int_{\pi/4}^{\pi/2} \frac{\sec^2 \theta d\theta}{\tan \theta \sqrt{1+\tan^2 \theta}}$$

$$= \int_{\pi/4}^{\pi/2} \frac{\sec^2 \theta d\theta}{\tan \theta \sec \theta}$$

$$= \int_{\pi/4}^{\pi/2} \frac{1}{\cos \theta} \frac{\cos \theta}{\sin \theta} d\theta$$

$$= \int_{\pi/4}^{\pi/2} \frac{1}{\sin \theta} d\theta$$

(ii) $I = \int_{\pi/4}^{\pi/2} \csc \theta d\theta$

$$I = \left[-\ln(\csc x + \cot x) \right]_{\pi/4}^{\pi/2}$$

$$= \left[\ln(\csc x + \cot x) \right]_{\pi/4}^{\pi/2}$$

$$= \ln\left(\csc \frac{\pi}{4} + \cot \frac{\pi}{4}\right) - \ln\left(\csc \frac{\pi}{2} + \cot \frac{\pi}{2}\right)$$

$$= \ln(\sqrt{2}+1) - \ln(1+0)$$

$$= \ln(\sqrt{2}+1)$$

$$\begin{aligned} \text{(c) (i)} \quad x^4 + x^3 + x^2 + x + 1 &= (x^2 + ax + 1)(x^2 + bx + 1) \\ &= x^4 + bx^3 + x^2 + ax^3 + abx^2 + ax + x^4 + bx + 1 \\ &= x^4 + x^3(a+b) + x^2(2+ab) + x(a+b) + 1 \end{aligned}$$

$$\therefore (a+b) = 1 \quad \dots (1)$$

$$2+ab = 1 \quad \dots (2)$$

$$ab = -1$$

$$b = \frac{-1}{a}$$

$$a - \frac{1}{a} = 1$$

$$a^2 - 1 = a$$

$$a^2 - a - 1 = 0$$

$$a = \frac{1 \pm \sqrt{1-4 \cdot (-1) \cdot 1}}{2}$$

$$= \frac{1 \pm \sqrt{5}}{2}$$

$$\therefore a = \frac{1+\sqrt{5}}{2}, \quad b = \frac{1-\sqrt{5}}{2} \quad [\text{by symmetry of equation}]$$

$$\text{(ii)} \quad \therefore \cos \frac{2\pi}{5} + i \sin \frac{2\pi}{5} \text{ is a soln of } x^2 + ax + 1 \text{ or } x^2 + bx + 1.$$

$$(\cos \frac{2\pi}{5} + i \sin \frac{2\pi}{5})^2 + a(\cos \frac{2\pi}{5} + i \sin \frac{2\pi}{5}) + 1 = 0$$

$$\cos \frac{4\pi}{5} + i \sin \frac{4\pi}{5} + a \cos \frac{2\pi}{5} + a i \sin \frac{2\pi}{5} = -1$$

equating real

$$\cos \frac{4\pi}{5} + a \cos \frac{2\pi}{5} = -1$$

$$\cos 2\theta = 2\cos^2 \theta - 1$$

$$2\cos^2 \frac{2\pi}{5} - 1 + a \cos \frac{2\pi}{5} = -1$$

$$\cos \frac{2\pi}{5} (2\cos \frac{2\pi}{5} + a) = 0$$

$$\therefore \cos \frac{2\pi}{5} = -\frac{a}{2} \quad (\text{or } -b)$$

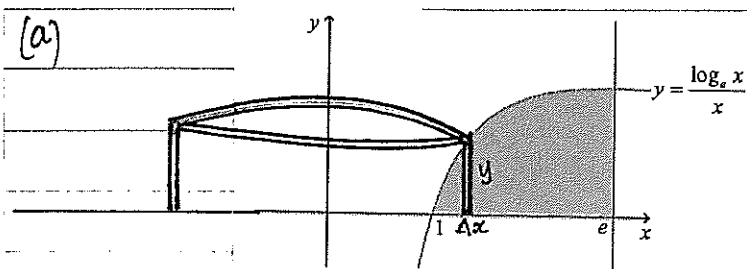
$$\cos \frac{2\pi}{5} = -\frac{1-\sqrt{5}}{4} \quad \text{or} \quad \cos \frac{2\pi}{5} = -\frac{1+\sqrt{5}}{4}$$

$$\text{But } \cos \frac{2\pi}{5} > 0$$

$$\therefore \cos \frac{2\pi}{5} = -\frac{1+\sqrt{5}}{4}$$

Question 6

(a)



$$\text{Area of Annulus} = \pi x^2 - \pi (x - \Delta x)^2$$

$$= \pi x^2 - \pi (x^2 - 2x\Delta x + (\Delta x)^2)$$

$$= 2\pi x\Delta x$$

↑
too small

$$\therefore \text{Volume of Shell} = 2\pi x\Delta x y$$

$$\therefore \text{Volume of Solid} = \lim_{\Delta x \rightarrow 0} \pi \sum_{i=1}^e 2xy\Delta x$$

$$= 2\pi \int_1^e xy dx$$

$$= 2\pi \int_1^e x \log_e x dx$$

$$= 2\pi \int_1^e \log_e x dx$$

$$= 2\pi \text{ II}$$

$$= 2\pi \times 1$$

$$= 2\pi \text{ units}^3$$

$$\text{II} = \int_1^e \ln x dx \quad \text{let } u = \ln x \quad u' = \frac{1}{x}$$

$$= [x \ln x]_1^e - \int_1^e x \times \frac{1}{x} dx$$

$$= e - [e - 1]$$

$$= 1$$

* Note also $\int_1^e \ln x dx = 1$ from definition of area under $\log_e x$ from 1 to e

(b) (i) RTS $\sin(\alpha+\beta) + \sin(\alpha-\beta) = 2 \sin \alpha \cos \beta$

$$\text{LHS} = \sin(\alpha+\beta) + \sin(\alpha-\beta)$$

$$= \sin \alpha \cos \beta + \cos \alpha \sin \beta + \sin \alpha \cos \beta - \cos \alpha \sin \beta$$

$$= 2 \sin \alpha \cos \beta$$

$$= \text{RHS as required.}$$

(ii) $\sin \theta + \sin 2\theta + \sin 3\theta + \sin 4\theta$

$$= \sin(2\theta - \theta) + \sin(3\theta - \theta) + \sin(2\theta + \theta) + \sin(3\theta + \theta)$$

$$= \sin(2\theta + \theta) + \sin(2\theta - \theta) + \sin(3\theta + \theta) + \sin(3\theta - \theta)$$

$$= 2 \sin 2\theta \cos \theta + 2 \sin 3\theta \cos \theta$$

$$= 2 \cos \theta (\sin 2\theta + \sin 3\theta)$$

$$= 2 \cos \theta \left(\sin\left(\frac{5\theta}{2} - \frac{\theta}{2}\right) + \sin\left(\frac{5\theta}{2} + \frac{\theta}{2}\right) \right)$$

$$= 2 \cos \theta \times 2 \sin \frac{5\theta}{2} \cos \frac{\theta}{2}$$

$$= 4 \cos \theta \sin \frac{5\theta}{2} \cos \frac{\theta}{2}$$

$$\therefore 4 \cos \theta \sin \frac{5\theta}{2} \cos \frac{\theta}{2} = 0 \quad \begin{matrix} 0 \leq \theta < 2\pi \\ 0 \leq \frac{\theta}{2} \leq \pi \\ 0 \leq \frac{5\theta}{2} \leq 5\pi \end{matrix}$$

When $\cos \theta = 0$

$$\theta = \frac{\pi}{2}, \frac{3\pi}{2}$$

$\cos \frac{\theta}{2} = 0$

$$\frac{\theta}{2} = \frac{\pi}{2}$$

$$\theta = \pi$$

$$\sin \frac{5\theta}{2} = 0$$

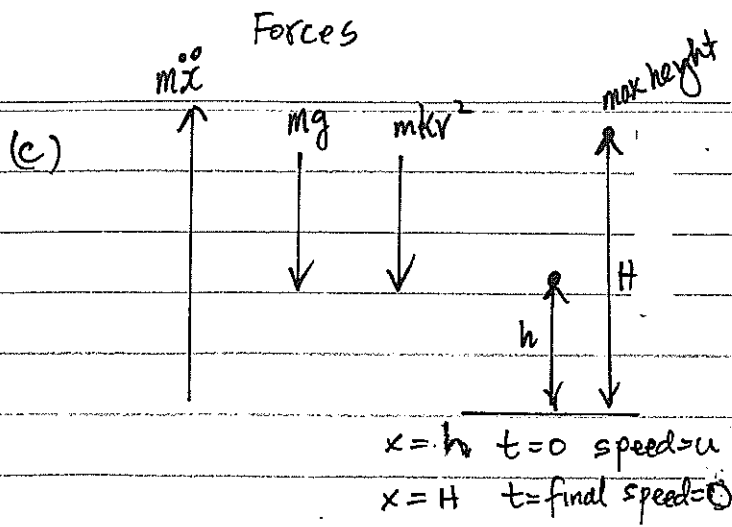
$$\frac{5\theta}{2} = 0, \pi, 2\pi, 3\pi, 4\pi, 5\pi$$

$$5\theta = 0, 2\pi, 4\pi, 6\pi, 8\pi, 10\pi$$

$$\theta = 0, \frac{2\pi}{5}, \frac{4\pi}{5}, \frac{6\pi}{5}, \frac{8\pi}{5}, 2\pi$$

$\therefore$ Solutions for θ are

$$0, \frac{2\pi}{5}, \frac{\pi}{2}, \frac{4\pi}{5}, \pi, \frac{6\pi}{5}, \frac{3\pi}{2}, \frac{8\pi}{5}, 2\pi$$



$$m\ddot{x} = -mg - mkv^2$$

$$\therefore \ddot{x} = -g - kv^2$$

$$v \frac{dv}{dx} = -g - kv^2$$

$$\frac{v dv}{-g - kv^2} = dx$$

$$-\int_u^0 \frac{v dv}{g + kv^2} = \int_h^H dx$$

$$\int_h^H dx = \int_0^u \frac{v}{g + kv^2} dv$$

$$\therefore H - h = \left[\frac{1}{2k} \ln(g + kv^2) \right]_0^u$$

$$= \frac{1}{2k} [\ln(g + ku^2) - \ln g]$$

$$= \frac{1}{2k} \left[\ln \frac{g + ku^2}{g} \right]$$

$$= \frac{1}{2k} \left[\ln \left(1 + \frac{ku^2}{g} \right) \right]$$

$$\therefore H = h + \frac{1}{2k} \ln \left(1 + \frac{ku^2}{g} \right)$$

as required

(d) 1 package rest
3 people together $n-3$ people
 $= n-3$ packages
 $\therefore 1 \text{ package} + n-3 \text{ package}$
 $= n-2 \text{ packages of people}$

no. of ways = $\frac{3! \times (n-2)!}{(n-2)!}$ ← divide since circle.
 3 particular people sit together.
 $= 3! (n-3)!$
 $= 6(n-3)!$

(e) RTS $n+2 {}^n C_r = {}^n C_r + 2 {}^n C_{r-1} + {}^n C_{r-2}$

LHS = $n+2 {}^n C_r$
 $= \frac{(n+2)!}{r!(n-r+2)!}$

RHS = ${}^n C_r + 2 {}^n C_{r-1} + {}^n C_{r-2}$
 $= \frac{n!}{r!(n-r)!} + 2 \frac{n!}{(r-1)!(n-r+1)!} + \frac{n!}{(r-2)!(n-r+2)!}$
 $= \frac{n!(n-r+1)(n-r+2) + 2n!(n-r+2)r + n!r(r-1)}{r!(n-r+2)!}$

$$= \frac{n! [n^2 - nr + 2n - nr + r^2 - 2r + 2 - r + n + 2rn - 2r^2 + 4r + r^2 - r]}{r!(n-r+2)!}$$

$$= \frac{n! [n^2 + 3n + 2]}{r!(n-r+2)!}$$

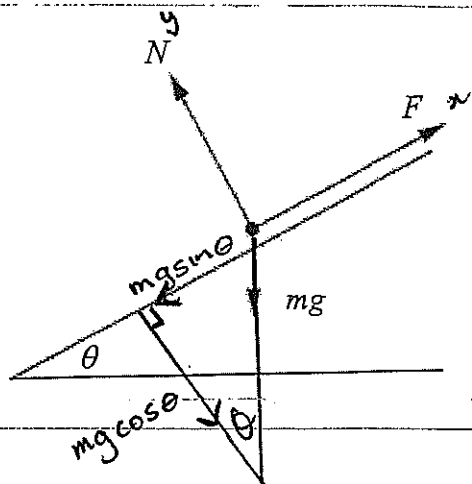
$$= \frac{n!(n+2)(n+1)}{r!(n-r+2)!}$$

$$= \frac{(n+2)!}{r!(n-r+2)!}$$

$$= \text{LHS as required}$$

Q7

(a)



$$F_x = F - mg \sin \theta \dots \textcircled{1} \text{ parallel}$$

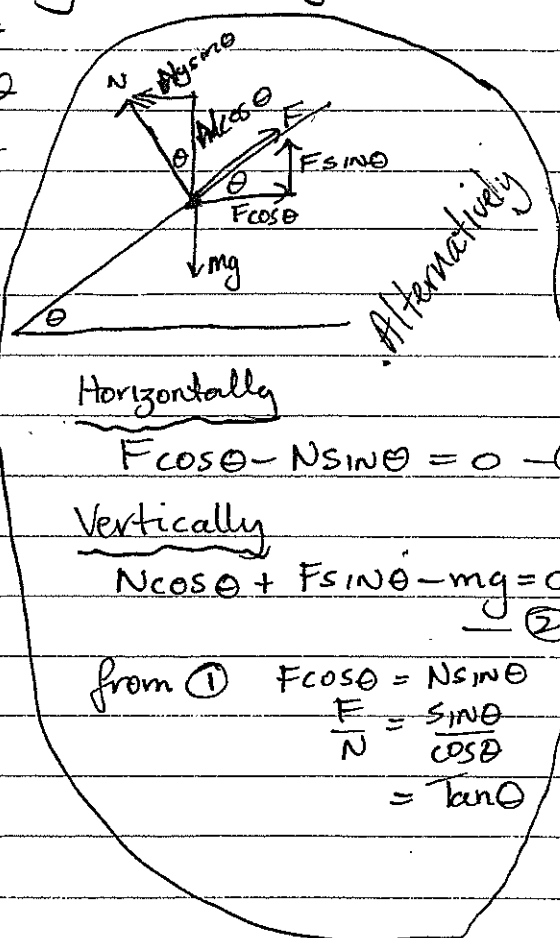
$$F_y = N - mg \cos \theta \dots \textcircled{2} \text{ perpendicular}$$

Since not moving $F_x = 0, F_y = 0$

$$\therefore F = mg \sin \theta$$

$$N = mg \cos \theta$$

$$\frac{F}{N} = \tan \theta$$



Horizontally

$$F \cos \theta - N \sin \theta = 0 \dots \textcircled{1}$$

Vertically

$$N \cos \theta + F \sin \theta - mg = 0 \dots \textcircled{2}$$

from $\textcircled{1}$ $F \cos \theta = N \sin \theta$

$$\frac{F}{N} = \frac{\sin \theta}{\cos \theta} = \tan \theta$$

(b) $P(x) = x^4 - 4x^3 + 3x^2 - 4x + 10$

(i) Roots $a+ib, a-ib, a-2ib, a+2ib$.

sum of roots

$$4a = \frac{-(-4)}{1}$$

$$= 4$$

$$a = 1.$$

Product of Roots:

$$(1+ib)(1-ib)(1-2ib)(1+2ib) = 10$$

$$(1+b^2)(1+4b^2) = 10$$

$$1+4b^2+b^2+4b^4 = 10$$

$$4b^4 + 5b^2 - 9 = 0$$

$$4b^4 + 9b^2 - 4b^2 - 9 = 0$$

$$b^2(4b^2+9) - (4b^2+9) = 0$$

$$(b^2-1)(4b^2+9) = 0$$

$$b^2 = 1 \quad [b \text{ real}]$$

$$b = \pm 1.$$

$$\left. \begin{array}{l} \alpha = 1+i \\ \bar{\alpha} = 1-i \end{array} \right\} \begin{array}{l} \alpha + \bar{\alpha} = 2 \\ \alpha \bar{\alpha} = 1^2 + 1^2 = 2 \end{array}$$

$$\left. \begin{array}{l} \beta = 1+2i \\ \bar{\beta} = 1-2i \end{array} \right\} \begin{array}{l} \beta + \bar{\beta} = 2 \\ \beta \bar{\beta} = 1^2 + 2^2 = 5 \end{array}$$

(ii) $(z-\alpha)(z-\bar{\alpha}) = z^2 - z(\alpha+\bar{\alpha}) + \alpha\bar{\alpha}$

$$(z-\beta)(z-\bar{\beta}) = z^2 - z(\beta+\bar{\beta}) + \beta\bar{\beta}$$

$\therefore (z^2 - 2z + 2)(z^2 - 2z + 5)$ is the factored form of $P(x)$ over the rational field.

Question 7

$$(c) (i) \quad I_n = \int_0^{\pi/2} \cos^n x \, dx$$

$$= \int_0^{\pi/2} \cos x \cos^{n-1} x \, dx$$

let $u = \cos^{n-1} x$ $v = \cos x$
 $u' = (n-1) \cos^{n-2} x \cdot (-\sin x)$ $v' = \sin x$

$$\therefore I_n = \left[\sin x \cos^{n-1} x \right]_0^{\pi/2}$$

$$- \int_0^{\pi/2} \sin x (n-1) \cos^{n-2} x \cdot (-\sin x) \, dx$$

$$= 0 + (n-1) \int_0^{\pi/2} \cos^{n-2} x \cdot \sin^2 x \, dx$$

$$= (n-1) \int_0^{\pi/2} \cos^{n-2} x (1 - \cos^2 x) \, dx$$

$$= (n-1) \int_0^{\pi/2} \cos^{n-2} x \, dx - (n-1) \int_0^{\pi/2} \cos^n x \, dx$$

$$= (n-1) I_{n-2} - (n-1) I_n$$

$$\therefore n I_n = (n-1) I_{n-2}$$

$$\therefore I_n = \frac{n-1}{n} I_{n-2}$$

as required.

$$(ii) \therefore \int_0^{\pi/2} \cos^5 x \, dx = I_5$$

$$= \frac{4}{5} I_3$$

$$= \frac{4}{5} \left(\frac{2}{3} I_1 \right)$$

$$= \frac{4}{5} \times \frac{2}{3} \times 1$$

$$= \frac{8}{15}$$

$$I_1 = \int_0^{\pi/2} \cos x \, dx$$

$$= \left[\sin x \right]_0^{\pi/2}$$

$$= 1$$

$$(d) \text{ RTP } 1 \times 1! + 2 \times 2! + 3 \times 3! + \dots + n \times n! = (n+1)! - 1$$

for $n \geq 1$

step 1 prove for $n=1$

$$\text{LHS} = 1 \times 1! = 1$$

$$\text{RHS} = (1+1)! - 1 = 2! - 1 = 1$$

$$\therefore \text{LHS} = \text{RHS} \therefore \text{true for } n=1$$

step 2 assume true for $n=k$

$$\text{i.e. } 1 \times 1! + 2 \times 2! + \dots + k \times k! = (k+1)! - 1$$

step 3 prove for $n=k+1$

$$\text{i.e. } 1 \times 1! + 2 \times 2! + \dots + (k+1) \times (k+1)! = [(k+1)+1]! - 1$$

$$\text{LHS} = 1 \times 1! + 2 \times 2! + \dots + k \times k! + (k+1) \times (k+1)!$$

$$= (k+1)! - 1 + (k+1)(k+1)!$$

$$= (k+1)! + (k+1)! (k+1) - 1$$

$$= (k+1)! [1 + (k+1)] - 1$$

$$= (k+1)! (k+2) - 1$$

$$= (k+2)! - 1$$

$$= [(k+1)+1]! - 1$$

$$= \text{RHS}$$

$\therefore$ result is true for $n=k+1$

Step 4 Since result is true for $n=1$,

it is also true for $n=2$. Since it is true for $n=2$ it is also true for $n=3$ and so on.

$\therefore$ Statement is true $\forall$ positive integral values of n .

Question 8

(a) $f(x) = \sqrt{3 - \sqrt{x}}$

(i)

Domain $3 - \sqrt{x} \geq 0$ and $\sqrt{x} \geq 0$
 $-\sqrt{x} \geq -3$ $x \geq 0$

$$\sqrt{x} \leq 3$$

$$x \leq 9$$

$$\therefore 0 \leq x \leq 9$$

(ii) $f(x) = (3 - x^{1/2})^{1/2}$

$$\therefore f'(x) = \frac{1}{2} (3 - x^{1/2})^{-1/2} \times -\frac{1}{2} x^{-1/2}$$

$$= \frac{-1}{4\sqrt{x}\sqrt{3 - \sqrt{x}}}$$

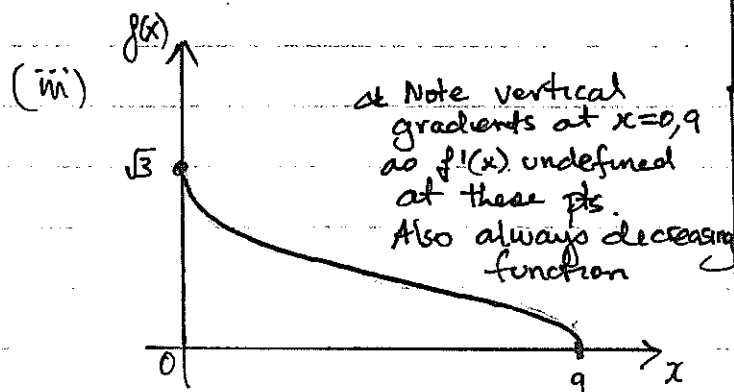
< 0 for $0 < x < 9$
 and grad. undefined for
 $x = 0$ and $x = 9$
 (ie. vertical)

$\therefore f(x)$ is a decreasing function.

at $x = 0$ $f(0) = \sqrt{3}$

at $x = 9$ $f(9) = 0$

$$\therefore 0 \leq f(x) \leq \sqrt{3}$$



(iv) $\int_0^9 \sqrt{3 - \sqrt{x}} dx = \int_3^0 \sqrt{u} (-2)(3 - u) du$

let $u = 3 - \sqrt{x}$

$$\sqrt{x} = 3 - u$$

$$x = (3 - u)^2$$

$$dx = -2(3 - u) du$$

$x = 0$ $u = 3$

$x = 9$ $u = 0$

$$= 2 \int_0^3 (3u^{1/2} - u^{3/2}) du$$

$$= 2 \left[2u^{3/2} - \frac{2u^{5/2}}{5} \right]_0^3$$

$$= 4 \left[3\sqrt{3} - \frac{9\sqrt{3}}{5} \right]$$

$$= 4\sqrt{3} \left[3 - \frac{9}{5} \right]$$

$$= 4\sqrt{3} \times \frac{6}{5}$$

$$= \frac{24\sqrt{3}}{5} \text{ as required.}$$

Question 8

b) RTS: $\frac{d}{du}(\sec u + \tan u) = \sec u(\sec u + \tan u)$

LHS = $\frac{d}{du}(\sec u + \tan u)$

= $\frac{d}{du}((\cos u)^{-1} + \tan u)$

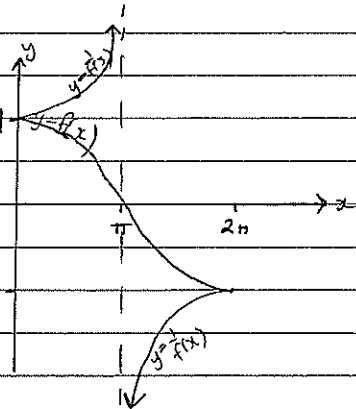
= $-(\cos u)^{-2} \times -\sin u + \sec^2 u$

= $\frac{\sin u}{\cos^2 u} + \sec^2 u$

= $\sec u \cdot \tan u + \sec^2 u$

= $\sec u(\sec u + \tan u)$

c) (i)



(ii) $A = \int_{\pi/3}^{\pi/2} \sec x \, dx$

Let $u = \frac{x}{2}$
 $\frac{du}{dx} = \frac{1}{2}$

= $2 \int_{\pi/6}^{\pi/4} \sec u \, du$ when $x = \pi/3$
 $u = \pi/6$

= $2 \int_{\pi/6}^{\pi/4} \frac{\sec u(\sec u + \tan u)}{\sec u + \tan u} du$ when $x = \pi/2$
 $u = \pi/4$

= $2 \left[\ln(\sec u + \tan u) \right]_{\pi/6}^{\pi/4}$

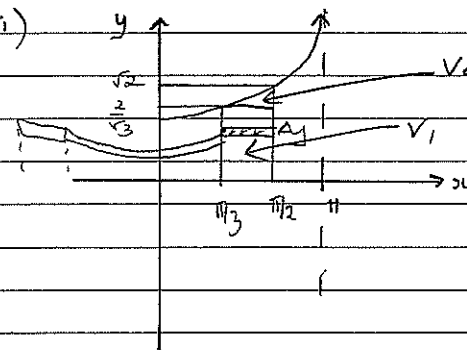
= $2 \left[\ln\left(\sec \frac{\pi}{4} + \tan \frac{\pi}{4}\right) - \ln\left(\sec \frac{\pi}{6} + \tan \frac{\pi}{6}\right) \right]$

= $2 \left[\ln(\sqrt{2} + 1) - \ln\left(\frac{2}{\sqrt{3}} + \frac{1}{\sqrt{3}}\right) \right]$

= $2 \left[\ln(\sqrt{2} + 1) - \ln\left(\frac{3}{\sqrt{3}}\right) \right]$

= $2 \ln\left(\frac{\sqrt{2} + 1}{\sqrt{3}}\right) \text{ units}^2$

(iii)



When $x = \pi/2$

$y = \frac{1}{\cos \pi/4}$

= $\frac{1}{\frac{1}{\sqrt{2}}}$

= $\sqrt{2}$

When $x = \pi/3$

$y = \frac{1}{\cos \pi/6}$

= $\frac{1}{\frac{\sqrt{3}}{2}}$

= $\frac{2}{\sqrt{3}}$

$V_1 = \pi \left(\left(\frac{\pi}{2} \right)^2 - \left(\frac{\pi}{3} \right)^2 \right) \times \frac{2}{\sqrt{3}}$

= $\pi \left(\frac{\pi^2}{4} - \frac{\pi^2}{9} \right) \times \frac{2}{\sqrt{3}}$

= $\pi \left(\frac{5\pi^2}{36} \right) \times \frac{2}{\sqrt{3}}$

= $\frac{10\pi^3}{36\sqrt{3}} \text{ units}^3$

$A_2 = \pi \left(\frac{\pi^2}{4} - x^2 \right)$

$V_2 = \sum_{y=\frac{2}{\sqrt{3}}}^{\sqrt{2}} \pi \left(\frac{\pi^2}{4} - x^2 \right) dy$

= $\pi \int_{\frac{2}{\sqrt{3}}}^{\sqrt{2}} \left(\frac{\pi^2}{4} - x^2 \right) dy$

But, $y = \frac{1}{\cos x}$

$$\frac{1}{y} = \cos \frac{x}{2}$$

$$\frac{x}{2} = \cos^{-1}\left(\frac{1}{y}\right)$$

$$x = 2 \cos^{-1}\left(\frac{1}{y}\right)$$

$$\therefore V_2 = \int_{\frac{2}{\sqrt{3}}}^{\sqrt{2}} \left(\frac{\pi^2}{4} - 4 \left[\cos^{-1}\left(\frac{1}{y}\right) \right]^2 \right) dy$$

$$\therefore V = \frac{10\pi^3}{36\sqrt{3}} + \int_{\frac{2}{\sqrt{3}}}^{\sqrt{2}} \left(\frac{\pi^2}{4} - 4 \left[\cos^{-1}\left(\frac{1}{y}\right) \right]^2 \right) dy$$